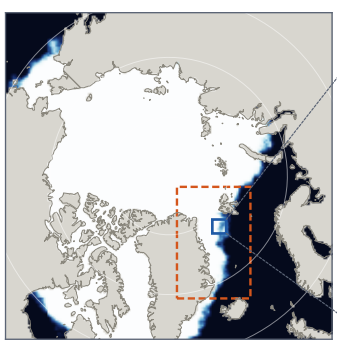
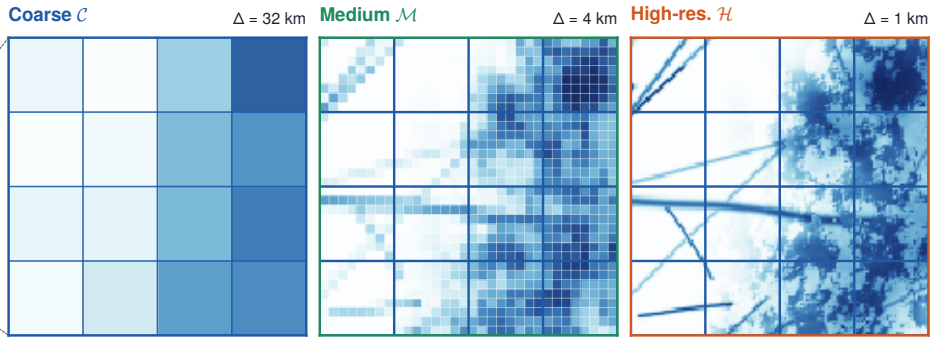
(a) Nested polar-stereographic grid: what each level resolves (nesting ×8, then ×4)

Arctic domain, $X_{\mathcal{C}}$



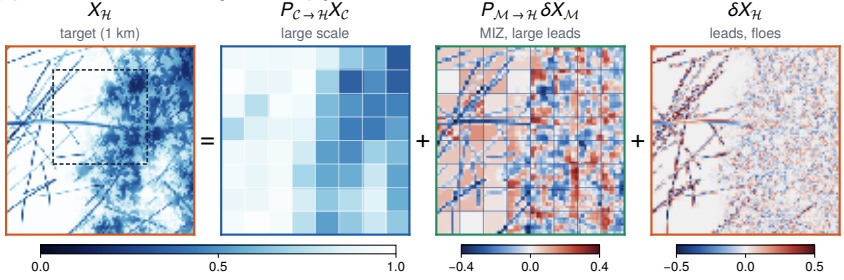
6400 km × 6400 km · 200×200 $\mathcal{C}$ cells
orange dashed: ROI where $\mathcal{H}$ is generated
(SAR coverage, illustrative)



4×4 cells: ice edge, pack mean
32×32 cells: MIZ, wide leads
128×128 cells: leads, floes

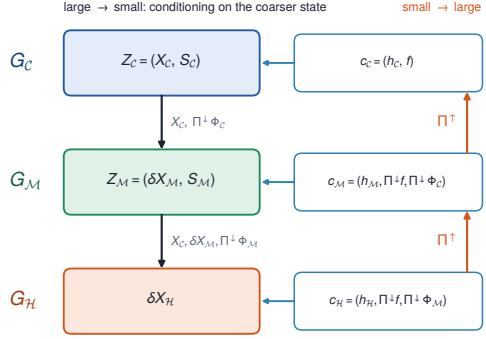
same 128-km window at the three levels ($X_{\mathcal{M}} = RX_{\mathcal{H}}$, $X_{\mathcal{C}} = RX_{\mathcal{M}}$); blue lines: $\mathcal{C}$ cells

(b) Exact scale decomposition (synthetic SIC, Fram Strait)



Restriction R (fixed)	Prolongation P (fixed)	Residual DOF
$X_{\mathcal{M}}(J) = \Sigma_{I\in \mathrm{ch}(J)}\,\bar{\omega}_I X_{\mathcal{H}}(I)$	$(PX_{\mathcal{M}})(i) = X_{\mathcal{M}}(\mathrm{par}(i))$	$\delta X_{\mathcal{M}} = X_{\mathcal{M}} - PX_{\mathcal{C}},\ \ R\,\delta X = 0$
ocean-weighted block mean	injection, $RP = I$	zero mean per parent

(c) Chain rule, coarse → fine



$$p_\theta(Z \mid Y, F) = p_\theta(Z_{\mathcal{C}} \mid c_{\mathcal{C}})\, p_\theta(Z_{\mathcal{M}} \mid Z_{\mathcal{C}}, c_{\mathcal{M}})\, p_\theta(\delta X_{\mathcal{H}} \mid Z_{\mathcal{C}}, Z_{\mathcal{M}}, c_{\mathcal{H}})$$

h_ℓ: obs. info (pyramid $\Pi^{\uparrow}$) · $\Phi_\ell = (S_\ell, \psi_\ell)$: sub-grid · f: forcing